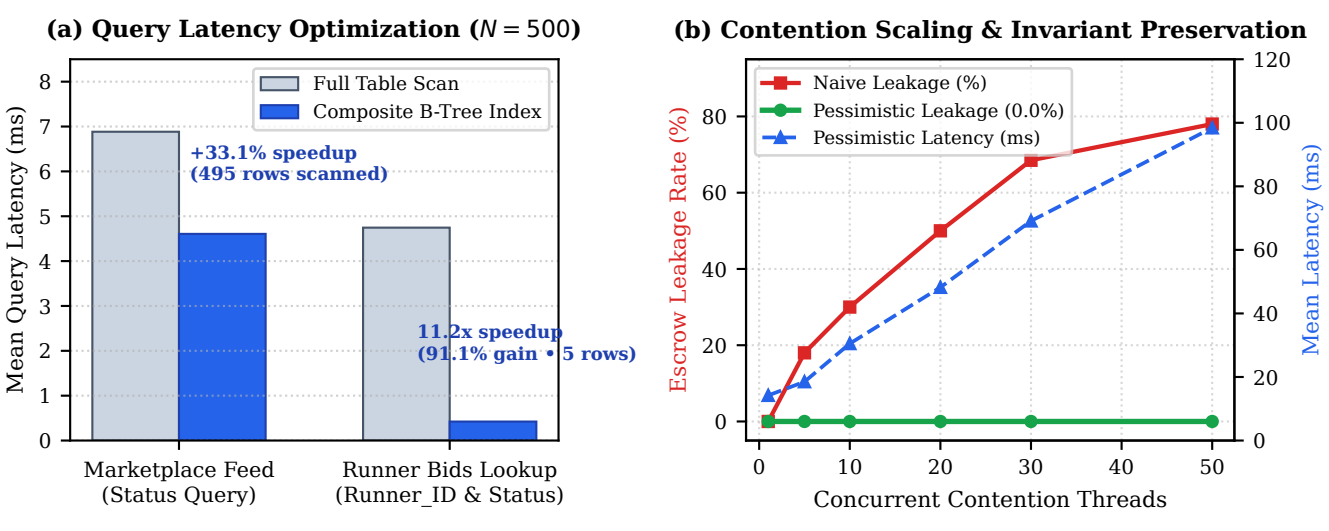
(a) Query Latency Optimization ($N = 500$)
Mean Query Latency (ms)
Full Table Scan
Composite B-Tree Index
+33.1% speedup
(495 rows scanned)
11.2x speedup
(91.1% gain • 5 rows)
Marketplace Feed
(Status Query)
Runner Bids Lookup
(Runner_ID & Status)
(b) Contention Scaling & Invariant Preservation
Escrow Leakage Rate (%)
Mean Latency (ms)
Naive Leakage (%)
Pessimistic Leakage (0.0%)
Pessimistic Latency (ms)
Concurrent Contention Threads